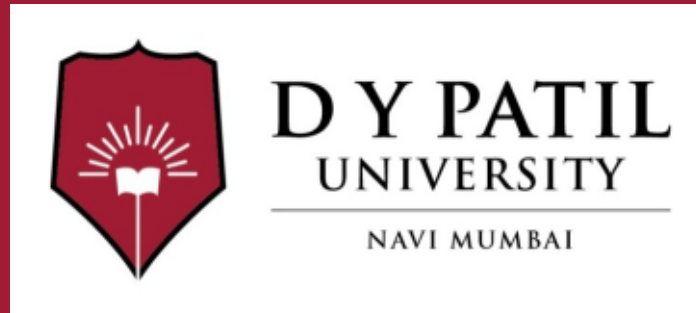




: Statistical Techniques

Unit 5 : Sampling and Hypothesis Testing

Faculty Name :Dr. Ashwini Kadam

Syllabus - Unit 5 : Sampling and Hypothesis Testing

-Sampling

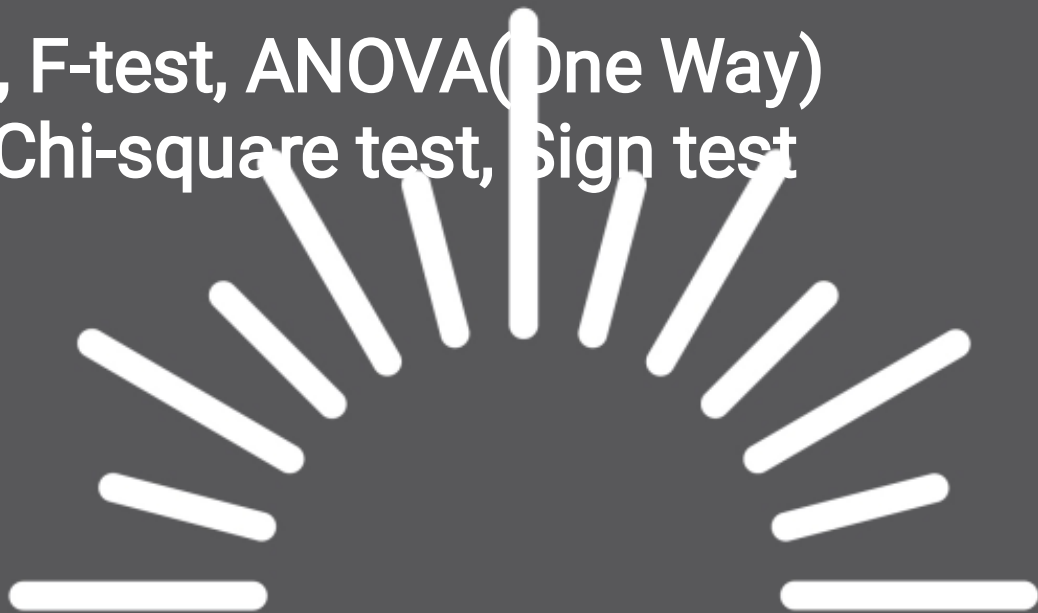
Population and Sample, Types of Population, Types of Sampling, Merits and Limitations of Sampling

-Hypothesis Testing

Meaning of Hypothesis, Null and Alternative Hypothesis, Type I and Type II errors, Level of Significance, Testing of Hypothesis

-Parametric test - Z-test, t-test, F-test, ANOVA(One Way)

-Non Parametric Test - Chi-square test, Sign test



Sampling-

-Population and Sample



Population

Population refers to the entire group of items, people, or observations that we want to study.

Example :

All users of a mobile app

All AI models developed in a company

All students using an e-learning platform

Example:

A company wants to analyze user satisfaction of its app.

Population = All 50,000 app users



Sample

A sample is a **subset of the population** selected for analysis.

A sample is a small group selected from a large population to represent the whole group in a study or research.

Example:

Surveying 500 users out of 50,000

Sample = 500 users

Purpose: Saves time, cost, and effort



Research Title:

“Impact of Coding Practice Hours on Machine Learning Model Development among MCA Students”

□ Population

All MCA students learning Machine Learning across different colleges.

Suppose:

Population = 50,000 MCA students

□ Sample

Instead of studying all 50,000 students, the researcher selects a smaller group.

Example:

Sample = 500 MCA students



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How Sample Works in This Research

The researcher collects data from these 500 students:

Hours spent on coding practice (X)

Number of ML models developed (Y)

Then uses this data to:

- ❑ Find patterns
- ❑ Perform analysis (correlation/regression)
- ❑ Draw conclusions for all students



Why Sample is Used

1. Saves Time

Studying 50,000 students is time-consuming
500 students → faster results

2. Cost Effective

Less data collection cost
No need for large surveys or infrastructure

3. Easy Data Handling

Research involves tools like Python, R
Smaller data → faster computation

4. Practical for AI/ML Projects

In real-world ML:

Models are trained on sample datasets, not entire population
Example: Training model on 10,000 records instead of millions



In fields like:

Data Science

Machine Learning

Software Analytics

You will always work with samples, because:

- ✓ Real-world data is too large
- ✓ Sampling makes analysis feasible and faster



Explain the importance of sampling in modern computer applications.

1. Efficient Handling of Large Data

Modern applications generate massive amounts of data every second.

Processing all of it is time-consuming and expensive.

-Sampling helps in working with a manageable subset of data.

Example:

A social media platform analyzes a sample of user posts to detect trends instead of scanning all posts.



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Explain the importance of sampling in modern computer applications

2. Faster Processing and Real-Time Performance

Applications like search engines, recommendation systems, and fraud detection require quick responses.

- Sampling reduces computation time and improves speed.

Example:

An online streaming platform uses sampled viewing data to recommend movies instantly.



Explain the importance of sampling in modern computer applications

3. Essential in Machine Learning Models

In Machine Learning, sampling is used to:

Split data into training and testing sets

Improve model accuracy

Avoid overfitting

Example:

A spam email filter is trained using a sample of emails rather than the entire email database.



Explain the importance of sampling in modern computer applications

4. Cost and Resource Optimization

Processing full datasets requires high memory, storage, and computing power.

-Sampling reduces hardware and operational costs.

Example:

A company analyzes a sample of transaction data instead of all transactions to reduce server load.



Explain the importance of sampling in modern computer applications

5. Software Testing Efficiency

Testing every possible case in software is not feasible.
-Sampling helps in selecting representative test cases.

Example:

A banking application tests selected user scenarios (login, transfer, withdrawal) instead of all combinations.



Explain the importance of sampling in modern computer applications

6. Better Decision-Making

Sampling provides quick insights, helping developers and managers take faster decisions.

Example:

A mobile app company collects feedback from a sample of users before launching a new feature.

Explain the importance of sampling in modern computer applications

7. Handling Big Data Analytics

In Big Data, datasets are extremely large and complex.
-Sampling enables real-time analytics and pattern detection.

Example:

An e-commerce platform studies sampled customer behavior to personalize product recommendations.

Explain the importance of sampling in modern computer applications

8. Supports Research and User Analysis

Computer applications often rely on surveys and usage data.

-Sampling ensures reliable insights without surveying every user.

Example:

A UI/UX team studies a sample of users to improve website design.

Conclusion

Sampling is essential in modern computer applications because it:

Saves time and cost

Improves processing speed

Enables efficient handling of big data

Supports accurate analysis and decision-making

-Without sampling, many modern systems would become slow, expensive, and inefficient.



Sampling-

-Types of Population



Types of Population

Population can be classified based on size, nature, and characteristics.

A. Based on Size

1. Finite Population
2. Infinite Population

B. Based on Characteristics

3. Homogeneous Population
4. Heterogeneous Population

C. Based on Existence

5. Real Population
6. Hypothetical (Theoretical) Population

D. Based on Time

7. Static Population
8. Dynamic Population



1. What is Population? (Quick Recap)

In statistics, population refers to the entire collection of elements (people, data, objects, events) about which we want to draw conclusions.

Examples :

All users of an application

All transactions in a database

All outputs generated by an AI model



Types of Population

Population can be classified based on size, nature, and characteristics.

A. Based on Size

1. Finite Population

Definition:

A population that has a fixed and countable number of elements.

Features:

- Clearly defined size
- Can be counted exactly
- Suitable for complete enumeration

Example:

- 300 students in MCA department
- 10,000 registered users in an app

Research Example:

Title: “Analysis of Coding Skills among MCA Students in School of Humanities and Sciences

Population = 300 students (finite)



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A. Based on Size

2. Infinite Population

Definition:

A population that has unlimited or uncountable number of elements.

Features:

Cannot be fully counted

Conceptual or continuous

Often theoretical in nature

MCA Example:

- Continuous server requests over time
- Data generated by IoT sensors
- Possible outcomes of AI model predictions

Research Example:

Title: “Real-Time Analysis of Website Traffic Patterns”

Population = Infinite clicks/visits over time



B. Based on Characteristics

3. Homogeneous Population

B. Based on Characteristics

3. Homogeneous Population

Definition:

A population where all elements are similar in characteristics.

Features:

Less variation

Easier to study

Smaller sample is sufficient

Example:

Students from same class with same syllabus

Same version software users

Research Example:

Title: “Performance of MCA Students in Python Programming”

All students have similar background



4. Heterogeneous Population

Definition:

A population where elements differ significantly.

Features:

- High variation
- Requires careful sampling (e.g., stratified sampling)
- More complex analysis

Example:

- App users from different countries, ages, skills
- Employees with different roles (developer, tester, manager)

Research Example:

Title: “User Satisfaction Analysis of a Global Mobile App”

Users differ in:

- Language
- Age
- Technical skills



C. Based on Existence

5. Real Population

5. Real Population

Meaning:

Population that actually exists and can be observed.

Key Points:

- Physically available
- Data can be collected directly

Example:

Employees in a company

Students in a university

Research Example:

👉 “Study of employee productivity in IT firms”



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C. Based on Existence

6. Hypothetical (Theoretical) Population

Meaning:

Population that is imaginary or conceptual.

Key Points:

Used in theory and probability

Not physically observable

Example:

All possible outputs of an AI model

All possible outcomes of a dice roll

Research Example:

“Simulation of AI model predictions”



D. Based on Time

7. Static Population

Meaning:

Population that remains constant over time.

Key Points:

- Does not change frequently
- Fixed during study period

Example:

- Students enrolled in a specific academic year
- Fixed dataset used for training ML model

•Research Example:

Analysis of final-year MCA students' project performance



Meaning:

Population that **changes over time.**

MCA Examples:

- Active users on a website
- Real-time stock market data
- Online gaming users

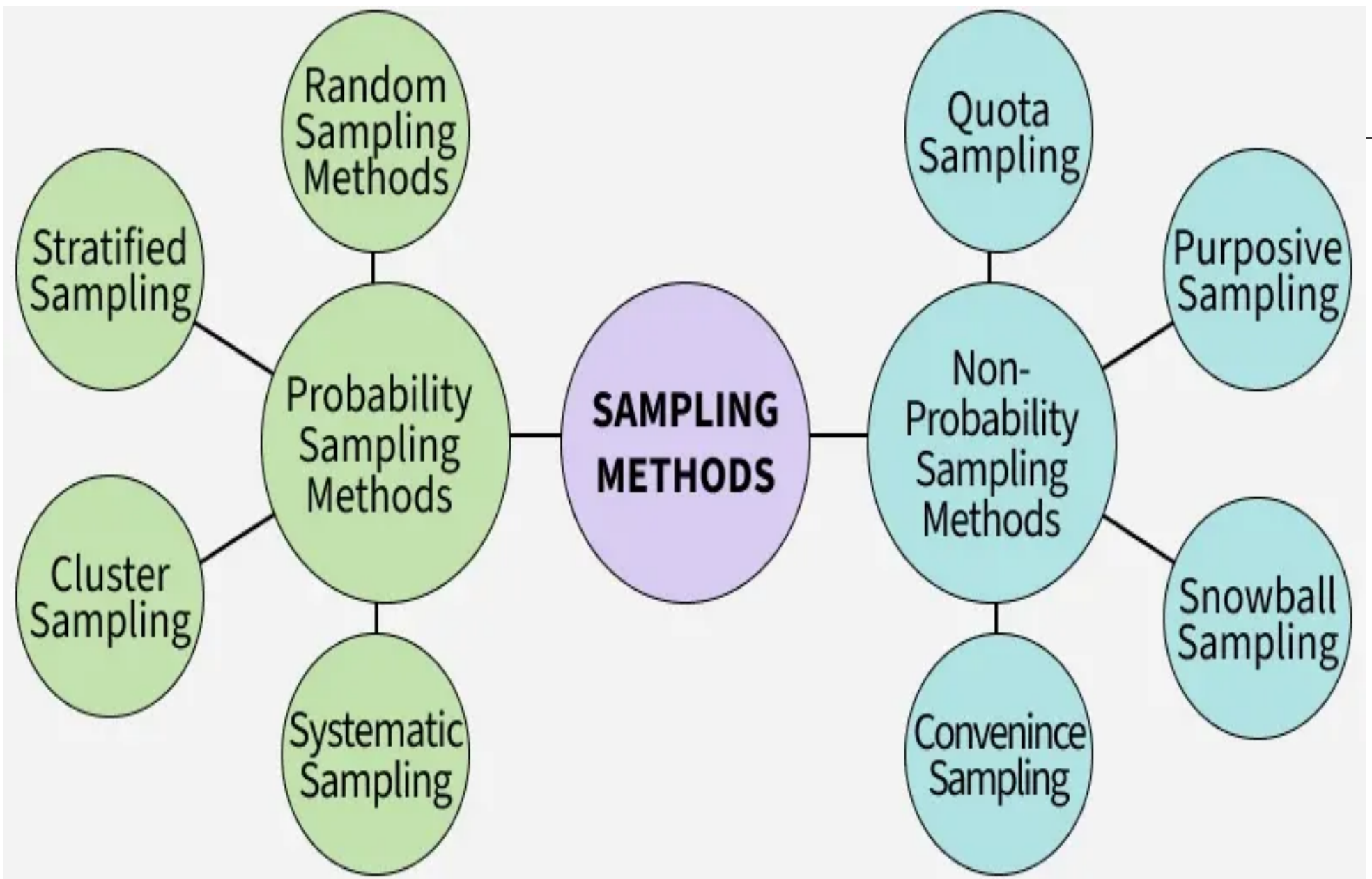
Research Example:

“Analysis of daily active users of a mobile application
”

Sampling-

-Types of Sampling





Types of Sampling

A. Probability Sampling (Scientific Method)

Each element in the population has a known and equal chance of being selected

Used in data science, AI research, surveys

B. Non-Probability Sampling (Non-Scientific)

Selection is based on judgment or convenience

Used when time or resources are limited



Probability Sampling

1. Simple Random Sampling

Meaning:

Every element has an **equal chance of selection**

How it works:

Lottery method

Random number generator

Example:

Selecting **100 users randomly** from a database of 10,000 users

Research Example:

“Study of coding skills among MCA students”

Randomly selecting students from list



Probability Sampling

2. Systematic Sampling

Meaning:

Selecting every nth element from a list

Formula:

Step size = Population / Sample

Example:

Selecting every 10th record from a dataset

Picking every 5th user from login data

Example:

Database of 1000 users

Sample = every 10th user → 100 users



Probability Sampling

3. Stratified Sampling

Meaning:

Population is divided into groups (strata), then samples are taken from each group

Why used:

To ensure representation of all categories

Example:

Dividing users into:

- Beginner
- Intermediate
- Advanced

Then selecting from each group

“Impact of Coding Practice Hours on Productivity of Software Developers in IT Companies”

Stratification:

Junior Developers

Mid-level Developers

Senior Developers

✓ Sample taken from each group → balanced analysis



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Cluster sampling

Cluster sampling is a method where the population is divided into groups (clusters), and then some clusters are randomly selected, and all individuals within those clusters are studied.

Cluster sampling is a sampling technique where:

The population is divided into clusters (groups)

Some clusters are randomly selected

All or some elements from selected clusters are studied

Key idea: *Sampling is done on groups, not directly on individuals*



Example-Cluster Sampling

Example 1: Software Companies

Population → All software engineers in India

Clusters → Companies (TCS, Infosys, Wipro)

Select random companies → Study all employees in them

Example 2: College Placement Study

Population → MCA students in India

Clusters → Colleges

Select some colleges → Study all students in them

Example 3: Software Companies

Population → All software engineers in India

Clusters → Companies (TCS, Infosys, Wipro)

Select random companies → Study all employees in them



Example-Cluster Sampling

Research Problem

A tech company wants to study:

“User satisfaction of a mobile app across India”

Step 1: Define Population

All app users in India

-Huge + geographically scattered → difficult to study individually

Step 2: Divide into Clusters

Divide users based on **cities**

Each city = one **cluster**

Example clusters:

Mumbai

Delhi

Bangalore

Pune

Hyderabad



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Non-Probability Sampling



Non Probability Sampling

Non-probability sampling is a sampling technique in which the selection of samples is non-random, the probability of selection of each unit is unknown, and the selection depends on the researcher's judgment, convenience, or network.

Key Features

- No randomization
- High chance of bias
- Quick and cost-effective
- Used in exploratory / qualitative research

TYPES OF NON-PROBABILITY SAMPLING

Convenience Sampling

Convenience sampling is a non-probability sampling technique in which the researcher selects participants who are easily available and accessible, without any random selection process, mainly to save time and cost.

Examples:

- A company surveys users currently active on its website
- Taking responses from people visiting a tech seminar



Convenience Sampling

Example

“Adoption of Cybersecurity Practices Among IT Professionals in Startups”

Explanation:

In this study, the researcher uses purposive sampling by selecting only those participants who have relevant expertise in cybersecurity, such as security analysts, ethical hackers, or experienced developers. The researcher intentionally chooses these individuals because they can provide meaningful insights related to the research topic. The selection is based on judgment rather than randomization, and not all IT professionals have a chance of being included. Hence, this is a non-probability sampling method, suitable for specialized research but limited in generalizing results to all professionals.



Purposive sampling

Purposive sampling is a non-probability sampling method in which the researcher deliberately selects participants based on their knowledge, expertise, or specific characteristics relevant to the study, so that the data collected is more meaningful and focused.

E.g.

In a study titled “*Performance Comparison of Deep Learning Models for Image Classification*”, the researcher selects only **AI/ML engineers and data scientists** because they have the required expertise in building and evaluating machine learning models.



Purposive sampling - Example

In a study titled “*Effectiveness of Machine Learning Algorithms in Predicting Stock Market Trends*”,

the researcher selects only data scientists and machine learning engineers because they have the required expertise in building and evaluating predictive models. The participants are deliberately chosen based on their technical knowledge and experience, ensuring that the data collected is relevant and insightful. Since the selection is based on judgment and not randomization, this study follows purposive (judgmental) sampling, which is suitable for specialized research but may limit generalization.



Snowball Sampling

Explanation:

Snowball sampling is a non-probability sampling technique in which existing participants refer or recruit additional participants, forming a chain that helps the researcher reach populations that are difficult to identify or access.

Research-based Example

In a study titled “*Freelancing Trends Among Remote Software Developers*”, the researcher initially contacts a few freelancers, who then refer other freelancers, expanding the sample size.

Additional Examples:

- Studying ethical hackers through referrals
- Research on startup founders via networking
- Finding participants in rare tech communities



Quota Sampling

Quota sampling is a non-probability sampling method in which the population is divided into **subgroups**, and a **fixed number of participants (quota)** is selected from each group, but the selection within each group is done non-randomly.

Examples:

- Selecting equal number of male and female app users
- Taking fixed number of front-end and back-end developers
- Choosing a set number of students from different semesters

Research-based Example

In a study titled “*User Satisfaction of Mobile Applications Based on Age Groups*”, the researcher selects a fixed number of users from different age groups (e.g., 50 users aged 18–25 and 50 users aged 26–40), but chooses them conveniently.



Basis	Probability Sampling	Non-Probability Sampling
Definition	Sampling in which each unit has a known and equal chance of selection	Sampling in which chance of selection is unknown
Selection Method	Random selection	Non-random selection
Bias	Less bias	High bias
Accuracy	More accurate and reliable	Less accurate
Generalization	Results can be generalized to population	Results cannot be generalized reliably
Time & Cost	Time-consuming and expensive	Quick and cost-effective
Examples	Simple random, stratified, cluster sampling	Convenience, purposive, quota, snowball sampling
Use Case	Large-scale, scientific research	Exploratory or preliminary research



Merits and Limitations of Sampling



Merits of Sampling

1. Saves Time

Sampling reduces the amount of data to be collected, so studies can be completed much faster than a full population study.

Example:

Testing a new app feature on 1,000 users instead of all users gives quick feedback.

2. Cost-Effective

It requires fewer resources like money, manpower, and tools compared to studying the entire population.

Example: Conducting a survey on selected customers instead of all users reduces survey cost.



Merits of Sampling

3. Faster Data Processing

Smaller datasets are easier and quicker to process using algorithms and software tools.

Example: Training a machine learning model on a sample dataset first to save computation time.

4. Feasibility

When it is impossible or impractical to study the whole population, sampling becomes the only workable option.

Example: Studying internet usage behavior of all users worldwide is impossible, so a sample is used.



Merits of Sampling

5. Better Manageability

A smaller dataset is easier to organize, clean, and analyze properly.

Example: Handling 10,000 database entries is much easier than managing millions of records.

6. Useful in Testing and Research

Sampling is widely used in experiments like A/B testing and system testing before full implementation.

Example: Testing two versions of a website on a small group of users before launching globally.



Merits of Sampling

7. High Accuracy (if properly done)

With proper sampling methods, results can closely represent the entire population.

Example: Stratified sampling ensures all user groups (age, region) are represented in analysis.



Limitations of Sampling

1. Sampling Bias

If the sample is not representative, it leads to incorrect conclusions about the population.

Example: Surveying only urban users for an app used in rural areas gives biased results.

2. Sampling Error

There is always a difference between sample results and actual population values.

Example: Average app usage time from a sample may differ from the true average of all users.

Limitations of Sampling

3. Requires Skilled Handling

Proper sampling design needs expertise; poor methods lead to unreliable results.

Example: Using convenience sampling instead of random sampling in a study can distort outcomes.

4. Risk of Overgeneralization

Results from a sample may not apply to the entire population in all cases.

Example: Testing a feature only on tech-savvy users may not reflect general user behavior.



Limitations of Sampling

5. May Miss Rare Data

Important but rare cases may not be included in the sample, affecting conclusions.

Example: Fraud detection systems may miss rare fraud patterns if sample size is small.

6. Sample Size Issues

Too small a sample reduces accuracy, while too large increases cost and effort.

Example: Using only 50 users to predict behavior of millions gives unreliable results.



Limitations of Sampling

7. Dependence on Sampling Technique

Different sampling methods can produce different outcomes, affecting consistency.

Example: Random sampling and cluster sampling may give different insights in user analytics.



Syllabus - Unit 5 : Sampling and Hypothesis Testing

-Hypothesis Testing

Meaning of Hypothesis, Null and Alternative Hypothesis, Type I and Type II errors, Level of Significance, Testing of Hypothesis

-Parametric test - Z-test, t-test, F-test, ANOVA(One Way)

-Non Parametric Test - Chi-square test, Sign test



Meaning of Hypothesis

A hypothesis is a tentative, logical, and testable statement about a population parameter or about the relationship between two or more variables. It is formulated before conducting research and is verified using statistical tests. A good hypothesis must be clear, specific, and capable of being tested using available data.

In research, the hypothesis provides direction and helps in designing experiments, collecting data, and drawing conclusions.

Example:

In a study on programming skills, a hypothesis may state that *increased coding practice improves software performance*.

Null Hypothesis (H_0)

The null hypothesis (H_0) is a statement that assumes there is no significant difference, no relationship, or no effect between the variables under study. It is the default assumption in statistical testing and is tested for possible rejection.

The null hypothesis is considered true unless sufficient evidence is found against it.

Example:

There is no relationship between coding practice hours and software performance.



Alternative Hypothesis (H_1 / H_a)

The alternative hypothesis (H_1) is a statement that contradicts the null hypothesis and suggests that there is a significant difference or relationship between variables. It represents the claim that the researcher wants to prove.

The alternative hypothesis is accepted only when the null hypothesis is rejected.

Example:

There is a significant improvement in software performance with increased coding practice.



Examples-Hypothesis

1. Academic Performance & Study Habits

H₀ (Null hypothesis): Study hours have no effect on MCA students' academic performance.

H₁ (Alternative hypothesis): Increased study hours improve MCA students' academic performance.

2. Programming Practice & Skill Development

H₀: Regular coding practice does not improve programming skills.

H₁: Regular coding practice improves programming skills.

3. Online Learning Platforms

H₀: Use of online learning platforms has no effect on students' performance.

H₁: Use of online learning platforms improves students' performance.



Examples-Hypothesis

4. Social Media & Productivity

H₀: Social media usage has no impact on academic productivity.

H₁: Excessive social media usage reduces academic productivity.

5. Use of AI Tools in Learning

H₀: AI tools do not affect learning outcomes of MCA students.

H₁: AI tools enhance learning outcomes of MCA students.

6. Internship Experience & Employability

H₀: Internship experience has no effect on employability.

H₁: Internship experience increases employability of MCA students.



Type I Error

A Type I error occurs when the null hypothesis is rejected even though it is actually true. In other words, it indicates detecting an effect or difference when none exists.

This error is also known as a false positive and is considered more serious in many research situations.

Example:

Concluding that coding practice improves performance when, in reality, it has no effect.



Type II Error

A Type II error occurs when the null hypothesis is not rejected even though it is false. This means failing to detect a real effect or difference.

This error is also known as a false negative.

Example:

Concluding that coding practice does not affect performance when it actually improves it.



1. Definition

The level of significance (α) is the maximum probability of committing a Type I error in hypothesis testing.

Type I error = Rejecting a true null hypothesis (H_0)

So,

$$\alpha = P(\text{Type I error})$$

Interpretation

- It represents the risk you are willing to take of being wrong.
- It is decided before collecting data.
- It sets the cutoff for rejecting H_0 .

Example:

$\alpha = 0.05 \rightarrow 5\%$ risk (5 out of 100 times you may wrongly reject H_0)

Common Values of α

α value	Meaning	Usage
0.10	10% error	Less strict
0.05	5% error	Most common
0.01	1% error	Very strict

Lower α = **more strict test**

Higher α = **more lenient test**



What is Confidence Level ?

The Probability of accepting the null Hypothesis (H_0) when the null hypothesis is true . It is denoted by $1 - \alpha$.

Relation with Level of Significance

$$\text{Confidence Level} = 1 - \alpha$$

α (Significance)	Confidence Level
0.10	90%
0.05	95%
0.01	99%

If confidence level = 95%

It means:

“If we repeat the experiment 100 times, about 95 times the interval will contain the true value.”



If confidence level = 95%

It means:

“If we repeat the experiment 100 times, about 95 times the interval will contain the true value.”



Exercise on Hypothesis formulation

Case study 1 : An MCA department introduces AI coding assistants in lab sessions. Faculty wants to check whether students using AI tools complete coding assignments faster than before. Students must decide:
Is there a real improvement in speed? Or is it just random variation?

Case Study 2 : A software engineering department recently introduced an AI-powered code auto-completion tool in programming labs. Faculty observed that students using this tool seem to finish assignments faster and with fewer syntax errors.

To evaluate this, data is collected from two groups:

Students using AI tools

Students not using AI tools

However, some teachers argue that the improvement might be due to practice over time, not the tool itself.

Students must decide:

Is the AI tool actually improving performance?

Or is the observed improvement just coincidental?



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Case Study 3: Group Study vs Self Study

Some students prefer group study sessions, discussing problems and solving coding tasks together. Others prefer studying alone.

Faculty wants to check whether:

Group study leads to significantly better exam performance

Or both methods are equally effective

Students must test:

Is there a difference in mean scores?

If yes, in which direction?



Case Study 4 : Attendance and Academic Performance

College policy emphasizes maintaining minimum 75% attendance, but faculty believes that students with attendance above 90% perform significantly better.

Students are grouped into:

High attendance (>90%)

Regular attendance (75–90%)

Students must analyze:

Is higher attendance actually linked to better performance?

Or is attendance not a significant factor?

Case Study 5: Internship and Placement Success

The placement cell claims that students who complete at least one internship during their course have better placement outcomes (higher package or selection rate).

Data is collected from:

Students with internships

Students without internships

Students must determine:

Is internship experience actually beneficial?

Or is placement success independent of internships?



Answers : Hypothesis Formulation

Case study 1

H₀: There is no improvement in coding speed after using AI tools.

H₁: There is improvement in coding speed after using AI tools.

Case study 2

H₀: There is no difference in performance between students using AI tools and those not using them.

H₁: There is a difference in performance between students using AI tools and those not using them.

Case Study 3

H₀: There is no difference in mean exam scores between students who study in groups and those who study alone.

H₁: There is a difference in mean exam scores between students who study in groups and those who study alone.



Answers : Hypothesis Formulation

Case study 4

H₀: There is no difference in performance between students with high attendance (>90%) and regular attendance (75–90%).

H₁: Students with high attendance (>90%) perform better than those with regular attendance (75–90%).

Case study 5

H₀: There is no difference in placement outcomes between students with internships and those without internships.

H₁: Students with internships have better placement outcomes than those without internships.

Syllabus - Unit 5 : Sampling and Hypothesis Testing

- Testing of Hypothesis
- Parametric test - Z-test



Z Test

What is a Z-test?

A Z-test is used to test whether a sample mean is significantly different from a population mean when population variance is known or sample size is large ($n \geq 30$).

In simple words:

“Is the difference real... or just by chance?”

When to Use Z-test?

Use Z-test when:

Sample size $n \geq 30$

Population standard deviation (σ) is known

Data is approximately normally distributed



Formula-Z test

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Where:

$\bar{X}$ = sample mean

μ = population mean

σ = population standard deviation

n = sample size



Z-value tells:

“How far is your sample from the population, measured in standard deviations?”

$Z = 0 \rightarrow$ perfect match

$Z = 2 \rightarrow$ far away

$Z = 5 \rightarrow$ something is definitely wrong



Hypothesis Setup

Step 1: Define Hypotheses

Null Hypothesis (H_0): No difference

Alternative Hypothesis (H_1): There is a difference

Example:

$$H_0: \mu = 50$$

$$H_1: \mu \neq 50$$



Level of Significance (α)

Usually **0.05 (5%)**

Means: 5% chance of rejecting a true hypothesis

Critical Z values:

$$\alpha = 0.05 \rightarrow \pm \mathbf{1.96}$$

$$\alpha = 0.01 \rightarrow \pm \mathbf{2.58}$$

Decision Rule

If $|\mathbf{Z}| \leq \mathbf{1.96} \rightarrow \text{Accept } H_0$

If $|\mathbf{Z}| > \mathbf{1.96} \rightarrow \text{Reject } H_0$



Example 1

A software training institute claims that the average coding test score of its students is 70 marks. Recently, a new teaching method using AI tools was introduced. To evaluate its effectiveness, a random sample of 64 MCA students who used the new method was selected. Their average score was found to be 74 marks. The population standard deviation is known to be 12 marks. Assume a 5% level of significance ($\alpha = 0.05$).

Questions :

1. Formulate the null hypothesis (H_0) and alternative hypothesis (H_1) to test whether the new teaching method has significantly improved student performance.
2. Based on the Z-test, decide whether: The new teaching method is effective
OR The observed improvement is just due to chance



Solution

Given Data

Population mean (μ) = 70

Sample mean ($\bar{X}$) = 74

Standard deviation (σ) = 12

Sample size (n) = 64

Significance level (α) = 0.05



Answer: Q1. : Hypothesis Formulation

Null Hypothesis (H_0)

H_0 : The new teaching method **does not improve** student performance

H_0 : $\mu = 70$ (No improvement)

Alternative Hypothesis (H_1)

H_1 : The new teaching method **improves** student performance

H_1 : $\mu > 70$ (Improvement due to new method)



Apply Z-Test Formula

$$Z = \frac{74-70}{12/\sqrt{64}}$$

$$Z = \frac{4}{12/8} = \frac{4}{1.5} = 2.67$$

Critical Value

- For $\alpha = 0.05$ (one-tailed) $\rightarrow Z_{\text{critical}} = 1.645$

Decision Rule

- Calculated $Z = 2.67$
- Critical $Z = 1.645$
- Since $2.67 > 1.645$, we reject H_0



Answer : Q2.

Q. 2 Based on the Z-test, decide whether: The new teaching method is effective OR
The observed improvement is just due to chance.

Answer :

There is statistically significant evidence that the new teaching method improves
student performance

The improvement is NOT due to chance



Example 2

A company advertises that its batteries last at least 500 hours on average. A testing agency checks 81 batteries and finds their average life to be 485 hours. The population variation is known to be 90 hours.

At a 5% level of significance, should the company's claim be accepted?

Questions:

Q. 1 : Formulate the null hypothesis (H_0) and alternative hypothesis (H_1) to test whether the company's claim about battery life is valid.

Q.2 Based on test result, decide whether: The company's claim that batteries last at least 500 hours is acceptable.



Answer Q1:

Since the company claims “at least 500 hours”, we test if the actual average is less than this value.

H₀: The average battery life is at least 500 hours

$\mu \geq 500$ hours (company’s claim is valid)

H₁: The average battery life is less than 500 hours

This is a left-tailed test

$\mu < 500$ hours (battery life is less than claimed)

Solution

Given:

Sample mean ($\bar{x}$) = 485

Population mean (μ) = 500

Population SD (σ) = 90

Sample size (n) = 81

Step 2: Z-value

$$Z = (485 - 500) / 10 = -15 / 10 = \mathbf{-1.5}$$

Step 3: Critical Value

At **5% significance level (one-tailed):**

Critical Z = **-1.645**



Solution

Step 4: Decision

Calculated $Z = -1.5$

Critical $Z = -1.645$

Since **$-1.5 > -1.645$** , we **fail to reject H_0** .



Example 3

An IT company claims that its developers write at least 200 lines of code per day on average. A sample of 64 developers shows an average of 210 lines per day. The population standard deviation is 40 lines. At a 5% level of significance, test whether the company's claim is valid.



Hypothesis

Null Hypothesis (H_0): The average number of lines of code written by developers is at least 200 lines per day.

Alternative Hypothesis (H_1): The average number of lines of code written by developers is less than 200 lines per day.



Given Data

Claimed mean $\mu=200$

Sample mean $\bar{X}=210$

Population standard deviation $\sigma=40$

Sample size $n=64$

Significance level $\alpha=0.05$



Solution

Step 2: Test Statistic

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$
$$Z = \frac{210 - 200}{40 / \sqrt{64}} = \frac{10}{5} = 2$$

Step 3: Critical Value

- At 5% level (left-tailed):

$$Z_{0.05} = -1.645$$



Syllabus - Unit 5 : Sampling and Hypothesis Testing

- Testing of Hypothesis
- Parametric test - t-test



Introduction

A **t-test** is a statistical test used to determine whether there is a significant difference between the means of two groups. It is widely used in research, data analysis, machine learning evaluation, and software analytics.

The t-test is an important statistical tool used to compare means and make data-driven decisions. It plays a major role in computer applications, research analysis, machine learning evaluation, and software performance testing.

The t-test is generally applied when:

- Sample size is small ($n < 30$)
- Population standard deviation is unknown
- Data follows approximately normal distribution



Example

1. Comparing Two Programming Languages

Problem:

An IT company wants to know whether Python developers complete coding tasks faster than Java developers.

Group 1: Python developers' completion time

Group 2: Java developers' completion time

Decision Making:

If the t-test shows significant difference, the company may choose Python for rapid development projects.

2 . Software Performance Optimization

Problem:

A software company compares execution time before and after code optimization.

Before optimization

After optimization

(Use Paired t-test)

Decision Making:

If improvement is statistically significant, optimized code will be deployed in production



Steps in Conducting a T-Test

1. Define Null and Alternative Hypothesis
2. Choose significance level ($\alpha = 0.05$ commonly)
3. Calculate t-value
4. Find degrees of freedom (df)
5. Compare calculated t-value with critical t-value
6. Draw conclusion



t-test Applications in computer applications

- Comparing algorithm performance
- Software testing analysis
- User behavior analysis
- Machine learning model evaluation
- Database performance comparison
- Comparing execution time of programs



Types of t-test

1. One-Sample T-Test

Used to compare the sample mean with a known or hypothesized population mean.

Example

Checking whether the average marks of MCA students differ from the university.

Formula

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = population mean
- s = sample standard deviation
- n = sample size

Types of t-test

2. Independent Sample T-Test

Used to compare means of two independent groups.

Example

Comparing marks of students taught using online learning vs classroom learning.

Formula

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{x}_1, \bar{x}_2$ = means of two groups
- s_1^2, s_2^2 = variances
- n_1, n_2 = sample sizes



3. Paired t-tests

Formula

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

Where:

- $\bar{d}$ = mean difference
- s_d = standard deviation of differences
- n = number of pairs



Degrees of Freedom

One Sample T-Test

$$Df = n - 1$$

Independent Sample T-Test

$$Df = n_1 + n_2 - 2$$

Paired T-Test

$$Df = n - 1$$



Decision Rule

If **$|t_{\text{calculated}}| > t_{\text{table}}$** $\rightarrow$ Reject H_0

If **$|t_{\text{calculated}}| \leq t_{\text{table}}$** $\rightarrow$ Accept (fail to reject) H_0



Syllabus - Unit 5 : Sampling and Hypothesis Testing

One Sample t-test



Decision Rule – t - test

Result

Decision

$$t < -t_{critical}$$

Reject H_0

$$t > +t_{critical}$$

Reject H_0

Between both values

Fail to reject H_0



Decision:

- If t is negative and large (more negative) $\rightarrow$ Reject H_0
- If t is positive $\rightarrow$ Do NOT reject H_0

Test Type	Does Sign Matter?	What to Check
Two-tailed	 No	
Right-tailed	 Yes	t positive
Left-tailed	 Yes	t negative



Example 1

An operating system claims that the average process execution time is 50 ms.

A sample of 16 processes shows an average time of 54 ms with a standard deviation of 8 ms.

Test at 5% level of significance whether the execution time is different.



Solution

Given Data

Population mean (μ) = 50 ms

Sample mean ($\bar{x}$) = 54 ms

Sample standard deviation (s) = 8 ms

Sample size (n) = 16

Level of significance = 5%

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$$t = \frac{54 - 50}{8 / \sqrt{16}} = \frac{4}{8/4} = \frac{4}{2} = 2$$



- Degrees of freedom: $df = n - 1 = 15$
 - At 5% (two-tailed), $t_{table} = 2.131$
-

•Calculated $t = 2$

•Table $t = 2.131$

Since $2 < 2.131$, we fail to reject H_0

There is no significant difference in execution time at the 5% level.

The operating system's claim (50 ms) is accepted.

Example 2

A training institute claims that students score an average of 75 marks in coding assessments. After introducing a new coding practice platform, a random sample of 10 students was selected. The sample showed an average score of 82 marks with a sample standard deviation of 8 marks. Test at 5% level of significance whether the new platform has improved the coding performance of students.



Solution

Hypothesis formulation

Null Hypothesis (H_0):

The new coding practice platform does not improve the coding performance of MCA students. The average coding assessment score remains 75 marks.

$$H_0: \mu = 75$$

Alternative Hypothesis (H_1):

The new coding practice platform improves the coding performance of MCA students. The average coding assessment score is greater than 75 marks.

$$H_1: \mu > 75$$



Calculation z value :

Use the one-sample t-test formula:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Substituting the values:

$$t = \frac{82 - 75}{8 / \sqrt{10}}$$

$$t = \frac{7}{2.53}$$

$$t \approx 2.77$$

$$t = \frac{7}{2.53}$$

$$t \approx 2.77$$

Degrees of freedom:

$$df = n - 1 = 9$$

Critical value at 5% significance level:

$$t_{critical} = 1.833$$

Since:

$$2.77 > 1.833$$

Reject the null hypothesis.

Conclusion:

There is sufficient evidence to conclude that the new coding platform improved the coding performance of MCA students.

Example 3

An MCA college introduced a new aptitude and coding bootcamp for final-year students to improve campus placements. After the training program, the placement cell selected a random sample of 15 students to analyze whether the average placement package increased beyond 6 LPA.

The sample showed an average placement package of 7.2 LPA with a sample standard deviation of 1.5 LPA.

Test at 5% level of significance whether the bootcamp improved the placement packages of MCA students.



Solution

Null Hypothesis (H_0):

The aptitude and coding bootcamp does not improve the placement packages of MCA students. The average placement package is 6 LPA.

$$H_0: \mu = 6 \text{ LPA}$$

Alternative Hypothesis (H_1):

The aptitude and coding bootcamp improves the placement packages of MCA students. The average placement package is greater than 6 LPA.

$$H_1: \mu > 6 \text{ LPA}$$



Solution

Given

$$\mu=6, \bar{x}=7.2, s=1.5, n=15$$

Formula

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Calculation

$$t = \frac{7.2 - 6}{1.5/\sqrt{15}}$$

$$t = \frac{1.2}{0.387}$$

$$t \approx 3.10$$

Solution

Critical Value

Degrees of freedom:

$df = n - 1 = 14$ At 5% significance level (right-tailed):

$t_{critical} = 1.761$

Decision

Since:

$3.10 > 1.761$ Reject the null hypothesis.

Conclusion

There is sufficient evidence to conclude that the aptitude and coding bootcamp significantly improved the placement packages of MCA students.



Example 4

After participating in a national hackathon, an MCA batch was evaluated to assess improvement in performance. The earlier average score of students was 70 marks. A random sample of 10 students from the batch showed an average score of 76 marks with a standard deviation of 8 marks. Test at the 5% level of significance whether the performance of students has improved after participating in the hackathon.



Solution

Formulate Hypothesis

Step 1: Hypotheses

H₀: The average score of students has not improved after the hackathon

H₀: $\mu \leq 70$ (no improvement)

H₁: The average score of students has improved after the hackathon

H₁: $\mu > 70$ (performance has improved)

This is a right-tailed test



Solution

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Given:

$$\bar{x} = 76$$

$$\mu = 70$$

$$s = 8$$

$$n = 10$$

Calculation:

$$\text{Standard Error (SE)} = 8 / \sqrt{10} \approx 8 / 3.16 \approx 2.53$$

$$t = (76 - 70) / 2.53 \approx 6 / 2.53 \approx \mathbf{2.37}$$



Solution

Step 3: Critical Value

Degrees of freedom (df) = $n - 1 = 9$

At 5% level (one-tailed), **t = 1.833**

Step 4: Decision

Calculated t = **2.37**

Critical t = **1.833**

Since **2.37 > 1.833**, we **reject H_0**

Conclusion

At the 5% level of significance, there is sufficient evidence to conclude that the performance of students has improved after participating in the hackathon.



Example 5

In a software company, the average time required to resolve customer support tickets is 40 minutes. After implementing a new AI-based support system, a sample of 9 support tickets shows an average resolution time of 34 minutes with a standard deviation of 6 minutes.

Test at the 5% level of significance whether the new system has reduced the resolution time.



Solution

Step 1: Hypotheses

Null Hypothesis (H_0):

The average time required to resolve customer support tickets is greater than or equal to 40 minutes, indicating that the new system has not reduced the resolution time.

$$H_0 : \mu \geq 40$$

Alternative Hypothesis (H_1):

The average time required to resolve customer support tickets is less than 40 minutes, indicating that the new system has reduced the resolution time.

$$H_1 : \mu < 40$$



Solution

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Given:

$$\bar{x} = 34$$

$$\mu = 40$$

$$s = 6$$

$$n = 9$$

Calculation:

$$SE = 6 / \sqrt{9} = 2$$

$$t = (34 - 40) / 2 = \mathbf{-3}$$



Solution

Step 3: Critical Value

$$df = 8$$

$$t(5\%, \text{one-tailed}) = -1.860$$

Step 4: Decision

Since $-3 < -1.860$, we **reject H_0**

Reason : This means calculated t is more negative than the critical value, so it lies in the rejection region.

Hence reject **H_0**

Step 5: Conclusion

There is sufficient evidence at the 5% level of significance to conclude that the new AI-based system has reduced the customer support resolution time.



Syllabus - Unit 5 : Sampling and Hypothesis Testing

-

Independent sample t-test



Independent Sample t-test

An independent t-test (also called a two-sample t-test) is a statistical method used to compare the means of two independent groups to determine whether there is a significant difference between them.

Use an independent t-test when:

- You have two separate groups (e.g., males vs females, treatment vs control)
- The data is continuous (e.g., height, scores, weight)
- You want to test if their means differ significantly



Independent Sample t-test

Formula

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

- $\bar{X}_1, \bar{X}_2$: sample means
- s_1^2, s_2^2 : sample variances
- n_1, n_2 : sample sizes

Example 1

An IT company compares productivity of developers using two different IDEs.

IDE (Integrated Development Environment) A group:
15 developers, average coding output = 220 lines/day,
SD = 30

IDE B (Integrated Development Environment) group:
12 developers, average coding output = 200 lines/day,
SD = 25

Test at 5% level of significance whether the choice of IDE affects developer productivity.

{At 5% level (two-tailed), $t_{table} \approx 2.06$ for $df=25$ }



Solution

Formulation of Hypothesis

$H_0: \mu_1 = \mu_2$ (no difference in productivity)

$H_1: \mu_1 \neq \mu_2$ (difference exists)

Null Hypothesis (H_0):

There is no significant difference in the average productivity of developers using IDE A and IDE B.

Alternative Hypothesis (H_1):

There is a significant difference in the average productivity of developers using IDE A and IDE B.



Solution

Given Data

- IDE A: $n_1 = 15$, $\bar{X}_1 = 220$, $s_1 = 30$
- IDE B: $n_2 = 12$, $\bar{X}_2 = 200$, $s_2 = 25$
- Level of significance: 5%



Solution

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$t = \frac{220 - 200}{\sqrt{\frac{30^2}{15} + \frac{25^2}{12}}}$$

$$t = \frac{20}{\sqrt{\frac{900}{15} + \frac{625}{12}}} = \frac{20}{\sqrt{60 + 52.08}} = \frac{20}{\sqrt{112.08}} = \frac{20}{10.59} = 1.89$$



Solution

$$Df = n_1 + n_2 - 2 = 15 + 12 - 2 = 25$$

Decision

Calculated $t = 1.89$

Table value = 2.06

Since $1.89 < 2.06$ Accept H_0

Final Conclusion

There is no statistically significant difference in developer productivity between IDE A and IDE B at the 5% level.

Interpretation

The choice of IDE does **not significantly affect productivity**, and the observed difference may be due to chance.



Example 2

An IT company compares developer productivity using two programming languages.

Language A (Python): 18 developers, mean output = 250 lines/day, SD = 35

Language B (Java): 20 developers, mean output = 230 lines/day, SD = 30

Test at 5% level whether the choice of programming language affects productivity.

{At 5% level (two-tailed), $t_{table} \approx 2.03$ for $df=36$ }



Solution

Hypothesis Formulation

H₀: There is no significant difference in productivity between Python and Java developers.

H₁: There is a significant difference in productivity between Python and Java developers.



Given Data

- Language A (Python): $n_1 = 18$, $\bar{X}_1 = 250$, $s_1 = 35$
- Language B (Java): $n_2 = 20$, $\bar{X}_2 = 230$, $s_2 = 30$
- Level of significance = 5%

$$t = \frac{250-230}{\sqrt{\frac{35^2}{18} + \frac{30^2}{20}}}$$

$$t = \frac{20}{\sqrt{\frac{1225}{18} + \frac{900}{20}}} = \frac{20}{\sqrt{68.06 + 45}} = \frac{20}{\sqrt{113.06}} = \frac{20}{10.63} = 1.88$$



$$df=n_1+n_2-2=18+20-2=36$$

Decision

Calculated $t=1.88$

Table value = 2.03

Since $1.88 < 2.03$, Accept H_0

There is no significant difference in developer productivity between Python and Java at the 5% level of significance.

Interpretation

The choice of programming language does not significantly affect productivity, and the observed difference may be due to random variation.



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Example 3

An IT company compares the effectiveness of two software testing tools in detecting bugs. A sample of 12 tests conducted using Tool A shows an average of 50 bugs detected with a standard deviation of 5. Another sample of 12 tests using Tool B shows an average of 40 bugs detected with a standard deviation of 6. Test at the 5% level of significance whether there is a significant difference in the bug detection capability of the two tools.

{At 5% level (two-tailed), $t_{table} \approx 2.07$ }



Hypothesis

H₀: There is no significant difference in bug detection capability between Tool A and Tool B

H₁: There is a significant difference in bug detection capability between Tool A and Tool B

$$t = \frac{50-40}{\sqrt{\frac{5^2}{12} + \frac{6^2}{12}}}$$

$$t = \frac{10}{\sqrt{\frac{25}{12} + \frac{36}{12}}} = \frac{10}{\sqrt{2.08 + 3}} = \frac{10}{\sqrt{5.08}} = \frac{10}{2.25} = 4.44$$



$$df = n_1 + n_2 - 2 = 12 + 12 - 2 = 22$$

Calculated $t = 4.44$

Table value = 2.07

Since $4.44 > 2.07$, we reject H_0

There is a statistically significant difference in the bug detection capability of the two tools.

Interpretation

The results indicate that the two software testing tools differ significantly in their effectiveness, and the observed difference is not due to chance.



Example 4

An IT firm compares the effectiveness of two antivirus systems in detecting threats. A sample of 10 tests conducted using System A shows an average of 55 threats detected with a standard deviation of 7.

Another sample of 12 tests using System B shows an average of 50 threats detected with a standard deviation of 6. Test at the 5% level of significance whether there is a significant difference in the detection capability of the two antivirus systems. ($t_{table} = 2.09$)



Hypothesis formulation

H₀: There is no significant difference in detection capability between System A and System B.

H₁: There is a significant difference in detection capability between System A and System B.

Solution

$$t = \frac{55-50}{\sqrt{\frac{7^2}{10} + \frac{6^2}{12}}}$$

$$t = \frac{5}{\sqrt{\frac{49}{10} + \frac{36}{12}}} = \frac{5}{\sqrt{4.9 + 3}} = \frac{5}{\sqrt{7.9}} = \frac{5}{2.81} = 1.78$$



$$df = n_1 + n_2 - 2 = 10 + 12 - 2 = 20$$

Calculated $t = 1.78$

Table value = 2.09

Since $1.78 < 2.09$, Accept H_0

There is no significant difference in the detection capability of the two antivirus systems at the 5% level of significance.

Interpretation :

The difference in the mean number of threats detected by System A and System B is not statistically significant, indicating that both systems perform similarly.



Syllabus - Unit 5 : Sampling and Hypothesis Testing

3. Paired t-tests



What is a Paired t-test?

Used when same subjects are measured twice (before–after, or two conditions on same group).

e.g. A group of MCA students is tested before and after a coding workshop to evaluate improvement in performance.



Paired t-test

- A paired t-test is a statistical method used to compare the means of two related samples to determine whether there is a significant difference between them.
- It is called *paired* because observations are taken from the same subjects at two different times or under two conditions.



When to Use Paired t-Test

Use this test when:

The same group is measured twice (Before–After)

Observations are dependent or matched

Example situations:

- Before vs After training
- System performance before vs after optimization
- Patient health before vs after treatment



Hypothesis

Null Hypothesis (H_0):

There is **no significant difference** between the two conditions
(*Mean difference = 0*)

Alternative Hypothesis (H_1):

There **is a significant difference**
(*Mean difference $\neq 0$ / > 0 / < 0 depending on test*)



Formula

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

Where:

- $\bar{d}$ = mean of differences
- s_d = standard deviation of differences
- n = number of pairs

(d = After – Before)

$$s_d = \sqrt{\frac{\sum (d - \bar{d})^2}{n - 1}}$$



Steps in Paired t-Test

- Compute differences: $d = X_1 - X_2$
- Find mean difference $\bar{d}$
- Calculate standard deviation sd
- Compute t-value
- Compare with critical t-value
- Draw conclusion



In exams, safest approach:

Define clearly:

$d = \text{Before} - \text{After}$

Or

$d = \text{After} - \text{Before}$

before calculations.

Then no marks are cut if hypotheses and decision are consistent.



Example 1

A software developer measures the execution time (in milliseconds) of a program before and after applying an optimization technique to determine whether the optimization reduces execution time. The execution times for 5 test cases are recorded as follows:

Test Case	Before (ms)	After (ms)
1	120	105
2	150	135
3	140	125
4	160	138
5	155	140

Question

Test at the 5% level of significance whether the optimization technique has significantly reduced the execution time of the program. Given that the tabulated t-value at 5% level of significance for 4 degrees of freedom is 2.776.



Solution

Hypothesis formulation

Null Hypothesis (H0)

The optimization technique does not significantly reduce the execution time.

Alternative Hypothesis (H1)

The optimization technique significantly reduces the execution time.



Solution

Calculate Mean Difference

$$\bar{d} = \frac{\sum d}{n}$$

$$\bar{d} = \frac{15 + 15 + 15 + 22 + 15}{5}$$

$$\bar{d} = \frac{82}{5} = 16.4$$



Solution

Test Case	Before (ms)	After (ms)	$d = \text{Before} - \text{After}$	$d - \bar{d}$	$(d - \bar{d})^2$
1	120	105	15	-1.4	1.96
2	150	135	15	-1.4	1.96
3	140	125	15	-1.4	1.96
4	160	138	22	5.6	31.36
5	155	140	15	-1.4	1.96



Solution

Calculate Standard Deviation

$$\sum (d - \bar{d})^2 = 39.2$$

$$s = \sqrt{\frac{\sum (d - \bar{d})^2}{n - 1}}$$

$$s = \sqrt{\frac{39.2}{4}}$$

$$s = \sqrt{9.8} = 3.13$$



Solution

Paired t-test formula:

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

Substituting values:

$$t = \frac{16.4}{3.13/\sqrt{5}}$$

$$t = \frac{16.4}{1.40}$$

$$t \approx 11.71$$

Decision

- Calculated t-value = 11.71
- Tabulated t-value = 2.776
- Degrees of freedom:

$$df = n - 1 = 5 - 1 = 4$$

Since:

$$11.71 > 2.776$$

Reject the null hypothesis.

Conclusion

At the 5% level of significance, the optimization technique significantly reduces the execution time of the program.



Example 2

A software company wants to determine whether a new debugging tool improves the bug-fixing performance of developers. The number of bugs fixed per day by 7 developers before and after using the tool is recorded as follows:

Developer	Before Tool	After Tool
1	18	24
2	20	27
3	22	30
4	19	25
5	21	29
6	17	23
7	24	32

Test at the 5% level of significance whether the new debugging tool significantly improves the bug-fixing performance of developers.

Given that the tabulated t-value at 5% level of significance for 6 degrees of freedom is 2.447.



Hypothesis Formulation

Null Hypothesis (H0)

The debugging tool does not significantly improve the bug-fixing performance of developers.

Alternative Hypothesis (H1)

The debugging tool significantly improves the bug-fixing performance of developers.

Solution

Calculate mean difference

$$\sum d = 49$$

$$\bar{d} = \frac{49}{7} = 7$$



Solution

Developer	Before Tool	After Tool	$d = \text{After} - \text{Before}$	$d - \bar{d}$	$(d - \bar{d})^2$
1	18	24	6	-1	1
2	20	27	7	0	0
3	22	30	8	1	1
4	19	25	6	-1	1
5	21	29	8	1	1
6	17	23	6	-1	1
7	24	32	8	1	1



Solution

Substitute in Formula

$$s = \sqrt{\frac{\sum (d - \bar{d})^2}{n - 1}}$$

Substituting values:

$$s = \sqrt{\frac{6}{7 - 1}}$$

$$s = \sqrt{\frac{6}{6}}$$

$$s = \sqrt{1}$$

$$s = 1$$



Solution

Calculate t-value

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

$$t = \frac{7}{1/\sqrt{7}}$$

$$t = \frac{7}{0.378}$$

$$t \approx 18.52$$



Solution

- Calculated t-value = 18.52
- Tabulated t-value = 2.447
- Degrees of freedom:

$$df = n - 1 = 7 - 1 = 6$$

Since:

$$18.52 > 2.447$$

Reject the null hypothesis.

Conclusion

At the 5% level of significance, the new debugging tool significantly improves the bug-fixing performance of developers.



Example 3

A company wants to determine whether a new HR management tool improves employee productivity. The number of tasks completed per day by 4 employees before and after using the HR tool is recorded as follows:

Employee	Before Using HR Tool	After Using HR Tool
1	15	20
2	18	24
3	20	27
4	16	22

Using a paired t-test, test at the 5% level of significance whether the HR management tool significantly improves employee productivity.

Given that the tabulated t-value at 5% level of significance for 3 degrees of freedom is 3.182.



Solution

Hypothesis Formulation

Null Hypothesis (H0)

The HR management tool does not significantly improve employee productivity.

Alternative Hypothesis (H1)

The HR management tool significantly improves employee productivity.



Solution

Since improvement means increase in productivity:

$d = \text{After} - \text{Before}$

Employee	Before	After	d	$d - \bar{d}$	$(d - \bar{d})^2$
1	15	20	5	-1	1
2	18	24	6	0	0
3	20	27	7	1	1
4	16	22	6	0	0

$$\sum d = 24$$

$$\sum (d - \bar{d})^2 = 2$$



Solution

$$\bar{d} = \frac{\sum d}{n}$$

$$\bar{d} = \frac{24}{4}$$

$$\bar{d} = 6$$



Solution

Standard Deviation

$$s = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}}$$

$$s = \sqrt{\frac{2}{3}}$$

$$s = \sqrt{0.667}$$

$$s \approx 0.816$$



Calculate t-value

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

$$t = \frac{6}{0.816/\sqrt{4}}$$

$$t = \frac{6}{0.408}$$

$$t \approx 14.71$$



-
- Calculated t-value = 14.71
 - Tabulated t-value = 3.182

Since:

$$14.71 > 3.182$$

Reject the null hypothesis.

Conclusion

At the 5% level of significance, the HR management tool significantly improves employee productivity.



Syllabus - Unit 5 : Sampling and Hypothesis Testing

F-TEST (Variance Ratio Test)



F-TEST (Variance Ratio Test)

The F-test is used to compare the variances of two populations to check whether they are equal.

It is based on the F-distribution.

Purpose of F-test

- To test equality of variances
- To check consistency or variability
- Used in ANOVA
- Helps in comparing performance stability



(A) Two Sample F-test

Compare variances of two groups

(B) F-test in ANOVA

Compare means of 3 or more groups

Based on variance ratio

Formula : F-test

$$F = \frac{\text{Larger Variance}}{\text{Smaller Variance}}$$

Degrees of Freedom

- $df1 = n1 - 1$
- $df2 = n2 - 1$



Steps in calculation of F-test

- State H_0 and H_1
- Calculate F-value
- Find degrees of freedom
- Give decision



Example 1

Two different algorithms are evaluated to study their consistency in execution time. A sample of 12 observations for Algorithm A shows a variance of 36, while a sample of 10 observations for Algorithm B shows a variance of 16.

At the 5% level of significance, test whether there is a significant difference in the variability of execution time between the two algorithms.

table value - At 5% level of significance:
3.10



Step 1: Hypothesis Formulation

Null Hypothesis (H_0):

There is no significant difference in the variability (variance) of execution time between Algorithm A and Algorithm B.

Alternative Hypothesis (H_1):

There is a significant difference in the variability (variance) of execution time between Algorithm A and Algorithm B.

Solution

Step 2: Given Data

Algorithm A

Variance = 36

Sample size (n_1) = 12

Algorithm B

Variance = 16

Sample size (n_2) = 10



Solution

Step 3: Calculate F-value

$$F = \frac{\text{Larger Variance}}{\text{Smaller Variance}}$$

$$F = \frac{36}{16} = 2.25$$



Step 4 : Degrees of Freedom

- $df_1 = n_1 - 1 = 12 - 1 = 11$
- $df_2 = n_2 - 1 = 10 - 1 = 9$

Table Value

At 5% level of significance:

$F_{0.05}(11,9) \approx 3.10$ (*approximate from F-table*)



Solution

Step 5: Decision

- $F_{cal} = 2.25$
- $F_{table} = 3.10$

Since $F_{cal} < F_{table}$, we accept H_0

Conclusion :

There is no significant difference in the variability of execution time between Algorithm A and Algorithm B. Hence, both algorithms have similar consistency.



Example 2

Two manufacturing machines are compared to study the consistency in the weight of items they produce.

A sample of 9 items from Machine A shows a variance of 81, while a sample of 7 items from Machine B shows a variance of 16.

At the 5% level of significance, test whether there is a significant difference in the variability of weights produced by the two machines.

At the 5% level of significance, the critical (table) value for degrees of freedom (8, 6) is 3.73



Solution

H_0 : There is no significant difference in variability of weights between the two machines.

H_1 : There is a significant difference in variability of weights between the two machines.

Given Data

Variance A = 81, $n_1 = 9$

Variance B = 16, $n_2 = 7$



Solution

$$F = 81/16 = 5.06$$

Degrees of Freedom

$$df_1 = 9 - 1 = 8$$

$$df_2 = 7 - 1 = 6$$

table Value

At 5% level:

$$F_{0.05}(8,6) \approx 3.73$$



Solution

$$F_{cal} = 5.06$$

$$F_{table} = 3.73$$

Since $F_{cal} > F_{table}$, we reject H_0

Decision

There is a significant difference in variability of weights produced by the two machines.

One machine is less consistent than the other.



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Syllabus - Unit 5 : Sampling and Hypothesis Testing

Testing of Hypothesis

-NON Parametric test – Chi-Square (χ^2) Test



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Testing of Hypothesis

-NON Parametric test – A) Chi-Square Goodness of Fit Test



Chi-Square (χ^2) Test

The Chi-Square Test is a non-parametric statistical test used to study the relationship between categorical variables.

It helps in:

- Testing whether observed data matches expected data
- Checking association between attributes
- Testing independence of variables

The symbol used is: χ^2



Types of Chi-Square Tests

A) Chi-Square Goodness of Fit Test

Used to check whether sample data fits a particular distribution.

Example

Checking whether customer preference for four mobile brands is equally distributed.

B) Chi-Square Test of Independence

Used to determine whether two categorical variables are related.

Example

Checking whether gender and programming language preference are associated.

Formula of Chi-Square Test

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where:

- O = Observed Frequency
- E = Expected Frequency

Steps in Chi-Square Test

Step 1: State Hypotheses

Null Hypothesis (H0):

No significant difference / variables are independent

Alternative Hypothesis (H1):

Significant difference / variables are dependent



Steps in Chi-Square Test

Step 2: Calculate Expected Frequencies

For independence test:

$$E = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$$

Step 3: Calculate Chi-Square Value

Use:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$



Steps in Chi-Square Test

Step 4: Find Degrees of Freedom

For Goodness of Fit Test

$$df = n - 1$$

Where:

- n = Number of categories

For Test of Independence

$$df = (r - 1)(c - 1)$$

Where:

- r = Number of rows
- c = Number of columns

Steps in Chi-Square Test

Step 5: Compare with Table Value

If calculated value $>$ table value $\rightarrow$ Reject H_0

Otherwise $\rightarrow$ Accept H_0



Example 1

A software company claims that customer preference for four web browsers is equally distributed. To verify this claim, a survey of 120 users was conducted and the following data was obtained:

Web Browser	Number of Users
Chrome	40
Firefox	30
Edge	25
Safari	25

Using the Chi-Square Test at 5% level of significance, test whether customer preference for web browsers is equally distributed.

Given: Level of significance $\alpha=0.05$, Table value of χ^2 for 3 degrees of freedom = 7.815



Solution

Step 1 :

Null Hypothesis (H0)

Customer preference for web browsers is equally distributed.

Alternative Hypothesis (H1)

Customer preference for web browsers is not equally distributed.

Explanation about hypothesis

In hypothesis testing, the Null Hypothesis (H0) is usually written as a statement of:

No difference

No effect

No association

Equal distribution

That is why here H0 is:

“Customer preference for web browsers is equally distributed.”

because the company is claiming that all browsers are preferred equally.

The Alternative Hypothesis (H1) is the opposite statement:

“Customer preference for web browsers is not equally distributed.”



Step 2: Calculate Expected Frequency

Since distribution is equal:

$$E = \frac{120}{4} = 30$$

So expected frequency for each browser = 30.



$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Browser	O	E	$O - E$	$(O - E)^2$	$\frac{(O-E)^2}{E}$	—
Chrome	40	30	10	100	3.33	
Firefox	30	30	0	0	0	
Edge	25	30	-5	25	0.83	
Safari	25	30	-5	25	0.83	

$$\chi^2 = 3.33 + 0 + 0.83 + 0.83$$

$$\chi^2 = 4.99$$



$$df=n-1$$

$$df=4-1=3$$

Calculated value:

$$\chi^2 = 4.99$$

Table value:

$$\chi^2_{0.05,3} = 7.815$$

Since:

$$4.99 < 7.815$$

Accept H_0 .



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Conclusion

There is no significant difference in customer preference for the web browsers.
Hence, customer preference for web browsers is equally distributed.



Example 2

A mobile store claims that customer preference for four smartphone brands is equally distributed. A survey of 200 customers produced the following results:

Brand	Customers
Samsung	60
Apple	50
OnePlus	45
Xiaomi	45

Using Chi-Square Test at 5% significance level, test whether customer preference is equally distributed.

Given:

$$\chi^2_{0.05,3} = 7.815$$

Solution

Total customers:

$$N=60+50+45+45=200$$

Number of categories:

$$n=4$$

Level of significance:

$$\alpha=0.05$$

Given table value:

$$\chi_{0.05,32}=7.815$$

Null Hypothesis (H₀)

Customer preference for smartphone brands is equally distributed.

Alternative Hypothesis (H₁)

Customer preference for smartphone brands is not equally distributed.



Since preference is equally distributed:

$$E = \frac{200}{4} = 50$$

Expected frequency for each brand = 50.



$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Brand	O	E	$O - E$	$(O - E)^2$	$\frac{(O-E)^2}{E}$	-
Samsung	60	50	10	100	2	
Apple	50	50	0	0	0	
OnePlus	45	50	-5	25	0.5	
Xiaomi	45	50	-5	25	0.5	

$$\chi^2 = 2 + 0 + 0.5 + 0.5$$

$$\chi^2 = 3$$



Solution

$$df=n-1$$

$$df=4-1=3$$

Calculated value:

$$\chi^2 = 3$$

Table value:

$$\chi_{0.05,3}^2 = 7.815$$

Since:

$$3 < 7.815$$

Accept H_0 .

Solution

Conclusion

There is no significant difference in customer preference for smartphone brands.
Hence, customer preference is equally distributed among the four brands.



Example 3

An MCA department believes students equally prefer four programming languages. A survey of 160 students gave the following data:

Language	Students
Java	55
Python	45
C++	35
JavaScript	25

Test whether preference is equally distributed at 5% significance level.

Given:

$$\chi^2_{0.05,3} = 7.815$$

Example 4

A company claims that website visitors come equally from four traffic sources. The observed data for 120 visitors is:

Traffic Source	Visitors
Search Engine	38
Social Media	32
Direct Visit	28
Advertisement	22

Test the claim using Chi-Square Test at 5% level.

Given:

$$\chi^2_{0.05,3} = 7.815$$

Syllabus - Unit 5 : Sampling and Hypothesis Testing

Testing of Hypothesis

-NON Parametric test – B) Chi-Square Test of Independence



Example 1

A software company wants to examine whether the type of application developed is independent of the programming language used.

The data collected is:

Application Type	Java	Python	C++	Total
Web Apps	30	20	10	60
Mobile Apps	20	25	15	60
Desktop Apps	10	15	25	50
Total	60	60	50	170

At 5% level of significance, test whether the two attributes are independent.

At 5% level of significance, test whether the two attributes are independent.

Given: χ^2 table value = 9.488 for df = 4



Step 1: Hypotheses

H_0 : Application type and programming language are independent

H_1 : They are not independent

$$E_{ij} = \frac{(\text{Row Total} \times \text{Column Total})}{\text{Grand Total}}$$

Calculations:

Row 1 (Web Apps): Total = 60

- Java: $(60 \times 60) / 170 = 21.18$
- Python: $(60 \times 60) / 170 = 21.18$
- C++: $(60 \times 50) / 170 = 17.65$



Solution

Row 2 (Mobile Apps): Total = 60

- Java: 21.18
 - Python: 21.18
 - C++: 17.65
-

Row 3 (Desktop Apps): Total = 50

- Java: $(50 \times 60) / 170 = 17.65$
- Python: $(50 \times 60) / 170 = 17.65$
- C++: $(50 \times 50) / 170 = 14.71$



Step 3: χ^2 Calculation Table

Cell	E	(O-E)	(O-E) ²	(O-E) ² / E
30	21.18	8.82	77.79	3.67
20	21.18	-1.18	1.39	0.07
10	17.65	-7.65	58.52	3.31
20	21.18	-1.18	1.39	0.07
25	21.18	3.82	14.59	0.69
15	17.65	-2.65	7.02	0.40
10	17.65	-7.65	58.52	3.31
15	17.65	-2.65	7.02	0.40
25	14.71	10.29	105.88	7.20



Solution

Total χ^2 Value

$$\chi^2 = 3.67 + 0.07 + 3.31 + 0.07 + 0.69 + 0.40 + 3.31 + 0.40 + 7.20$$

$$\chi^2 = 19.12$$



Solution

Step 4: Degrees of Freedom

$$df = (r - 1)(c - 1) = (3 - 1)(3 - 1) = 4$$

Step 5: Decision Rule

- Calculated $\chi^2 = 19.12$
- Table value = 9.488

Since:

$$19.12 > 9.488$$

Reject H_0



Final Conclusion

There is a significant association between application type and programming language.

Hence, the two attributes are not independent.



Syllabus - Unit 5 : Sampling and Hypothesis Testing

Testing of Hypothesis

-NON Parametric test – SIGN TEST



Sign test

Definition

The Sign Test is a non-parametric test used to test hypotheses about the median of a population.

Sign test compares median using $+/-$ signs and rejects H_0 when S is very small.

When to Use

Data is small sample ($n < 25$)

Data is ordinal / not normally distributed

Testing median, not mean

Used when t-test assumptions fail



One-sample sign test → compare median with a value

Paired sign test → before-after comparison

SIGN test Calculation steps

1. Set Hypothesis

H_0 : Median = M

H_1 : Median $\neq$ M / $>$ M / $<$ M

2. Assign Signs

Value $>$ M $\rightarrow$ +

Value $<$ M $\rightarrow$ -

Value = M $\rightarrow$ **Ignore**

3. Count Signs

Count number of + and -

Take **S = smaller of (+, -)**



Solution

4. Compare with Table

Use critical value (S table)

Decision Rule:

If $S \leq \text{critical value} \rightarrow \text{Reject } H_0$

If $S > \text{critical value} \rightarrow \text{Accept } H_0$

Formula

$$S = \min(n+, n-)$$



Key Point

The rule “**calculated > table** $\Rightarrow$ **reject H_0** ” is true for:

t-test

z-test

chi-square

F-test

BUT **✗ NOT for Sign Test**

✓ In Sign Test, the logic is REVERSED

Here:

Test statistic = **S = smaller of (+, -)**

Critical value is **very small (like 0, 1, 2)**



Reject H_0 if $S \leq \text{critical value}$

Accept H_0 if $S > \text{critical value}$



Example 1

A software company claims that the median number of coding tasks completed per day by employees is 15. A project manager believes the median productivity has increased after introducing a new project management tool. The numbers of tasks completed by 6 employees are given below:

Employee	Task Completed
1	18
2	12
3	17
4	20
5	14
6	16

Using the One Sample Sign Test at 5% significance level, test whether the median number of tasks completed is greater than 15. Given: Critical (table) value for $n=6$ at 5% significance level = 0.



Step 1: Set Hypothesis

Null Hypothesis (H0)

The median number of coding tasks completed per day by employees is 15 (i.e., there is no increase in productivity).

Alternative Hypothesis (H1)

The median number of coding tasks completed per day by employees is greater than 15 (i.e., productivity has increased after introducing the new tool).

Where M is the population median.



Step 2: Assign Signs

Compare each value with 15:

Value	Sign
18	+
12	-
17	+
20	+
14	-
16	+

Number of + = 4

Number of - = 2



Solution

Step 3: Test Statistic

$S = \text{smaller of } (+, -) = 2$

Step 4: Decision Rule

Given critical value = **0**

Compare:

$$S = 2 > 0$$

Accept H_0

Since calculated $S (2) > \text{critical value } (0)$, H_0 is accepted. Hence, there is no significant increase in median productivity.



Example 2

A software company claims that the median time to fix bugs is 10 hours. After introducing an AI debugging tool, a manager believes the fixing time has reduced. A sample of 7 bug-fixing times (in hours) is:

8, 9, 11, 7, 10, 6, 9

Question:

Using the One Sample Sign Test at 5% significance level, test whether the median bug-fixing time is less than 10 hours.

(Given: Critical value for $n = 7$, one-tailed test at 5% level = 1)



Solution

Compare each observation with median (10)

Observation	Value	Sign
1	8	-
2	9	-
3	11	+
4	7	-
5	10	0 (ignore)
6	6	-
7	9	-



Count signs

Positive signs (+) = 1

Negative signs (–) = 5

Ignore ties (0)

Effective sample size:

$n=6$

Test statistic S = smaller of (+, –) = **1**

Given: Critical value at 5% level (one-tailed, $n = 7$) = **1**



Example 3

A mobile app company claims that the median development time is **30 days**. A new agile method is introduced. A sample of 8 projects shows the following development times (days):
28, 32, 27, 29, 31, 26, 25, 30

Question:

Test whether the new method has **reduced** the median development time using the Sign Test at 5% significance.



Example 4

An IT firm claims that employees complete a median of **20 coding tasks per day**. After training, a manager believes productivity has increased. A sample of 9 employees shows:

22, 18, 21, 25, 19, 23, 20, 24, 26

Question:

Test whether the median productivity is **greater than 20** at 5% level using the Sign Test.



Example 5

A company claims that the median server response time is **200 ms**. After system optimization, a sample of 6 observations is recorded:

180, 190, 210, 170, 195, 185

Question:

Test whether the median response time is **less than 200 ms** using the Sign Test at 5% significance.

(Given: Critical value for $n = 6 = 0$)



Example 2 – Paired Sign Test

An institute introduces an AI-based learning tool to improve student performance. The marks obtained by 6 students before and after using the tool are as follows:

Before: 60, 65, 68, 62, 66, 64

After: 70, 63, 75, 69, 68, 66

Using the **paired sign test** at the 5% level of significance, test whether the AI-based tool has **improved student performance**.

Given: Critical value for $n=6$ ($n=6$) (one-tailed test at 5%) = **0**.



Hypotheses

H0: No improvement in performance

H1: Performance has improved

Step 2: Compute differences (After – Before)

Student	Before	After	Difference	Sign
1	60	70	+10	+
2	65	63	-2	-
3	68	75	+7	+
4	62	69	+7	+
5	66	68	+2	+
6	64	66	+2	+



Step 3: Count signs

Positive (+) = 5

Negative (−) = 1

$n=6$

Test statistic:

$S = \text{smaller of } (+, -) = \mathbf{1}$



Step 5: Decision Rule

Reject H_0 if $S \leq 0$

Here:

$$S = 1 > 0$$

Do not reject H_0



Example 3

An MCA faculty wants to check whether a new coding practice improves students' programming scores. The scores of 8 students before and after training are given below:



Student	Before Training	After Training
1	45	50
2	52	55
3	48	46
4	60	65
5	55	58
6	50	49
7	62	68
8	58	60



Question

Using the paired sign test, test at 5% level of significance whether the training improved performance.

Given: Critical value of Sign Test for $n=8$ at 5% level of significance (one-tailed test) is $S=1$.



Solution

Step 1: Hypotheses

H0: There is no improvement in scores.

H1: The training improves scores.



Step 2: Find Differences and Signs

$$d = \text{After} - \text{Before}$$

Student	Before	After	Difference d	Sign
1	45	50	+5	+
2	52	55	+3	+
3	48	46	-2	-
4	60	65	+5	+
5	55	58	+3	+
6	50	49	-1	-
7	62	68	+6	+
8	58	60	+2	+



Step 3: Count Signs

Positive signs = 6

Negative signs = 2

$n=8$ $S=\text{smaller value}=2$ $S = 2$



Step 4: Decision

Given table value:

$$S_{table} = 1$$

Decision rule:

- Reject H_0 if $S \leq S_{table}$

Here,

$$2 > 1$$

Therefore, Do not reject H_0 .



Solution

Conclusion

There is not enough evidence at 5% level of significance to conclude that the coding practice significantly improved students' programming scores.



Thank You



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